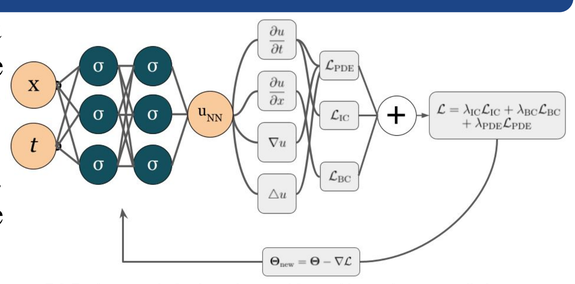
Maximum Entropy Regularization for Gradient Pathology Mitigation in Physics-Informed Neural PDE Solvers

Ishan Jha
Troy High School, Fullerton, CA.

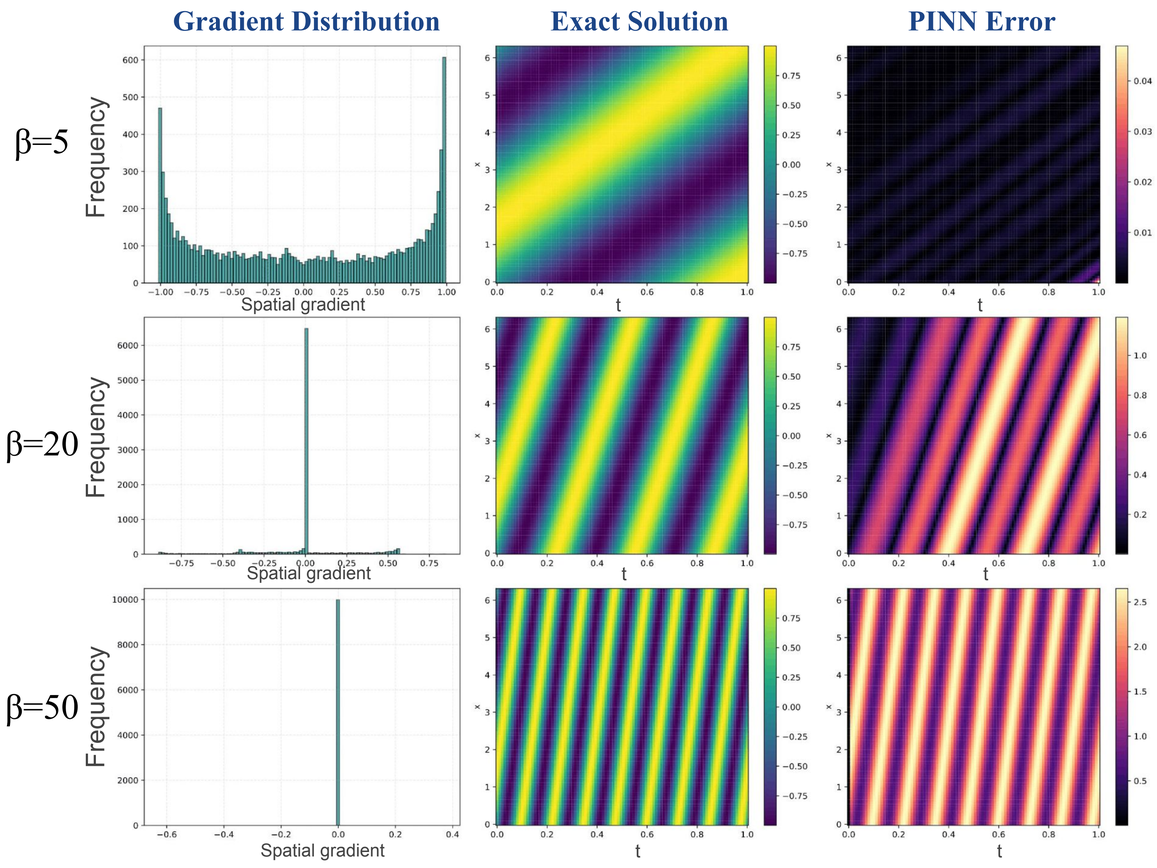
INTRODUCTION

Physics-informed neural networks (PINNs) [1] are a type of neural network that take advantage of the universal approximation property [2] to **numerically approximate** the solutions to partial differential equations (PDEs), yielding a differentiable solution. PINNs accomplish this through the modification of the loss function to include physics constraints.



- Prior work has documented **gradient pathologies** [3] in which the distribution of loss function gradients displays high concentration.
- In this work, we empirically show that this concentration extends to the gradient distribution of the PINN-*predicted solution.*
- We introduce **Gradient Entropy Regularization (GER)**, using the principle of maximum entropy as a framework, to penalize low gradient diversity in PINN solutions.
- We benchmark GER on a convection problem with non-trivial coefficient values and report L^2 error results in comparison with PINN.

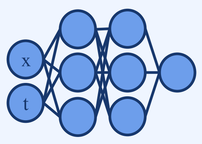
NUMERICAL PATHOLOGY



Given the following PDE:

$$\mathcal{D}[\mathbf{u}](x,t) = 0, \quad (x,t) \in \Omega \times [0,T]$$
$$\mathcal{B}[\mathbf{u}](x,t) = 0, \quad (x,t) \in \partial\Omega \times [0,T]$$
$$\mathbf{u}(x,0) = \mathbf{g}(x), \quad x \in \Omega$$

ALGORITHM SKETCH (GER)



(MLP-PINN)

$$\mathbf{u}(x,t) \approx \mathbf{u}_\theta(x,t)$$

(Surrogate)

Gradient

$$g^{(i)} = \nabla \mathbf{u}_\theta(x^{(i)}, t^{(i)}) \quad \mathcal{U}(\Omega \times [0,T])$$

Mean

$$\mu_k = \frac{1}{N_H}\sum_i g_k^{(i)}$$

Covariance

$$\sum_{i=1}^{N_H}(g_j^{(i)} - \mu_j)(g_k^{(i)} - \mu_k)$$

$$\widehat{\mathcal{P}_G} \approx \mathcal{N}(x; \boldsymbol{\mu}, \boldsymbol{\Sigma})$$

Entropy

$$\mathbb{E}[-\log \mathcal{N}]$$
$$H(G) = \frac{d}{2}\log(2\pi e) + \frac{1}{2}\log(\det \boldsymbol{\Sigma})$$

Loss

$$\mathcal{L} = \mathcal{L}_{PDE} + \mathcal{L}_{IC} + \mathcal{L}_{BC} + \mathcal{L}_H$$
$$\mathcal{L}_H = -H(G)$$

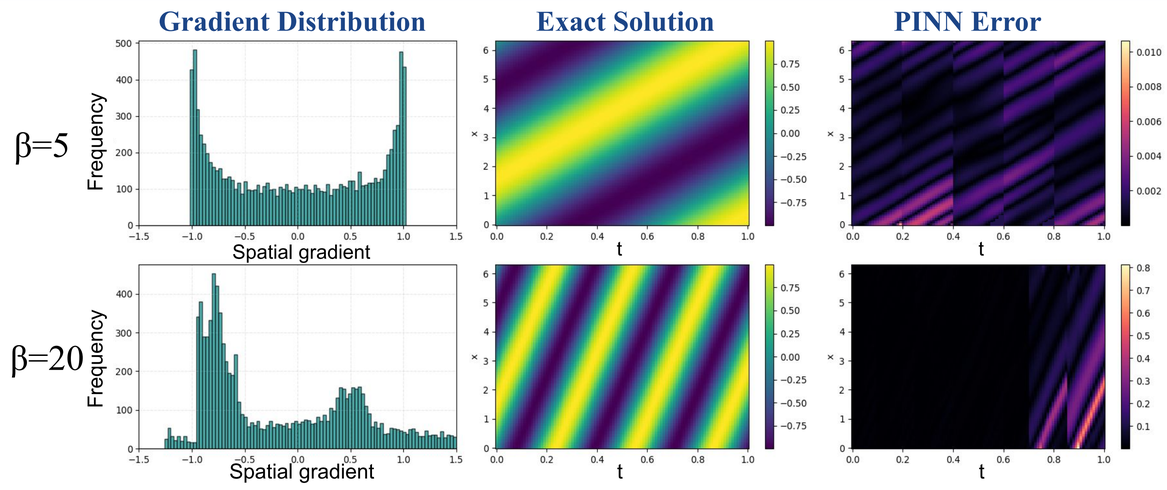
GER RESULTS

1D Convection

$$\frac{\partial u}{\partial t} + \beta\frac{\partial u}{\partial x} = 0$$
$$u(x,0) = \sin(x)$$
$$u(x,2\pi) = u(x,0)$$

Model Parameters

GER and PINN were evaluated on a convection PDE, where traditional PINNs are known to fail due to spectral bias. GER and PINN were both trained on 3,500 uniformly sampled points, with GER using 500 for entropy. Both models used 2000 ADAM [4] and 1000 L-BFGS iterations. Both models had a width of 64 and depth of 5. The weight for GER was set to 10^{-3}.



MODEL PERFORMANCE

Name	L^2 Error, β=5	L^2 Error, β=20	Points
GER	**0.12%**	**12.16%**	3,500 + 500 (GER)
PINN	0.14%	63.10%	3,500 (Total)

ANALYSIS

In this work, we develop and characterize a **maximum entropy** (MaxEnt) approach for **regularizing** PINNs and preventing common gradient pathologies associated with convection (transport) PDEs.

- PINN and GER appear to learn the solution with **similar accuracy** with low values of β, but GER significantly outperforms PINN in β=20. This can be attributed to the ability of GER to penalize solutions with low gradient diversity.
- We utilize a differentiable **Gaussian** model for the empirical distribution of gradient values, which yields an upper-bound value for the entropy. This is due to the property of the Gaussian distribution that uniquely maximizes entropy over **R** [5].
- GER, being a tanh network, faces **representational bottlenecks** in high-β problems. The error improvement of GER from β=5 to 20 shows that, past this bottleneck, GER would perform well on higher-β problems.
- As seen in solution plots, GER is able to reduce the error distribution from a global area of high error to a smaller region of low error.

FUTURE WORK

- The analysis and development of GER would benefit from rigorous **gradient analysis**, including descriptions of the behavior of the GER term during gradient descent.
  - This includes analysis of the gradients of the GER term within the total loss.
  - In an effort to make GER more **compatible** with loss terms, analysis of the influence ratio of GER relative to PDE and IC losses could be done.
- Investigation of **adaptive** weighting and residual-adaptive refinement (RAR).
- Implementation of **time-marching** (temporal/Euler discretization) and causal-weights for respecting the principle of causality in time-dependent problems.
- Further testing outside transport regimes in nonlinear conservation laws (Burgers equation), reaction (Allen-Cahn) dynamics and fluid dynamics problems.

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